Por convención, para calcular la volatilidad implícita de las opciones, utilizaremos la fórmula de Black / Black & Scholes.

$$C = S_t N(d_1) - K e^{-rT} N(d_2) = C(S_t, K, r, T, \sigma)$$

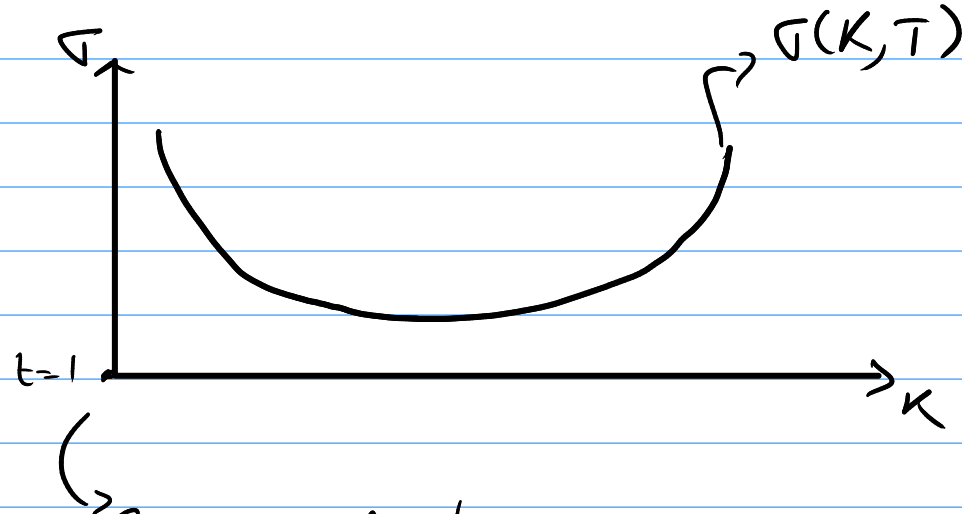
$$d_1 = \frac{\ln\left(\frac{S_t}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}} \qquad C \longleftrightarrow \sigma$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Problemas B&S → Asume σ constante para cualquier strike

$$dS_t = rS_t dt + \sigma S_t dW_t$$

En mercado

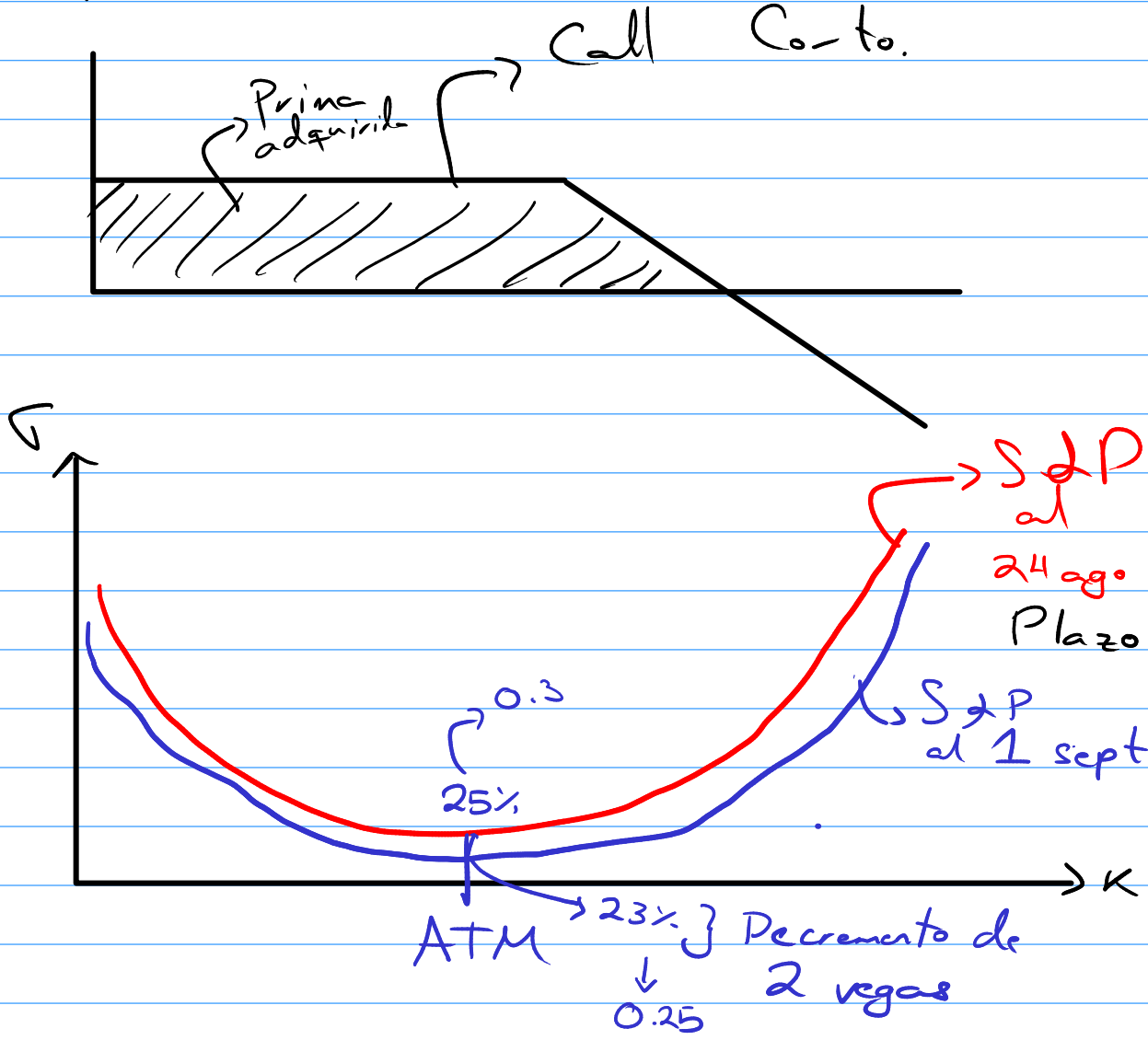


Causa de la existencia de un smile:
→ Las opciones tienen una dinámica "propia"

Michael Burry } "The Big Short"

↳ Short sobre posiciones del S&P y el Nasdaq.

→ Salió al mercado a vender opciones Call.

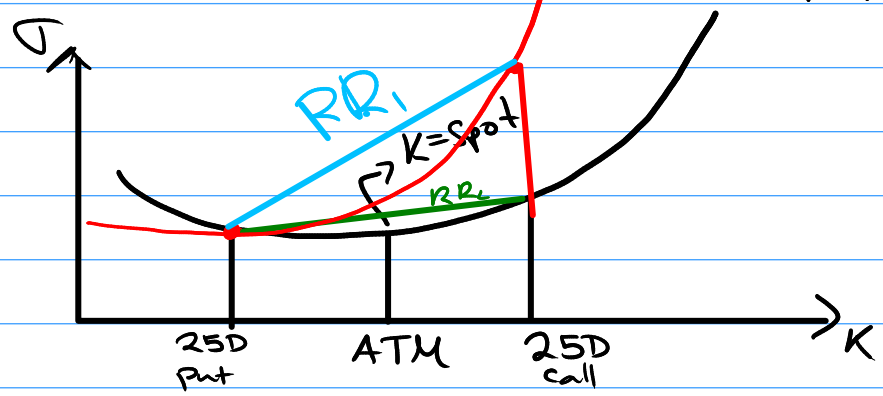


En el mercado FX, cotizan los siguientes instrumentos:

→ Risk Reversal
→ Butterfly

* $RR = \sigma_{25D\ call} - \sigma_{25D\ put}$

$RR_1 > RR_2$



↳ El strike K_1 tal que $\Delta C(K_1) = 0.75$

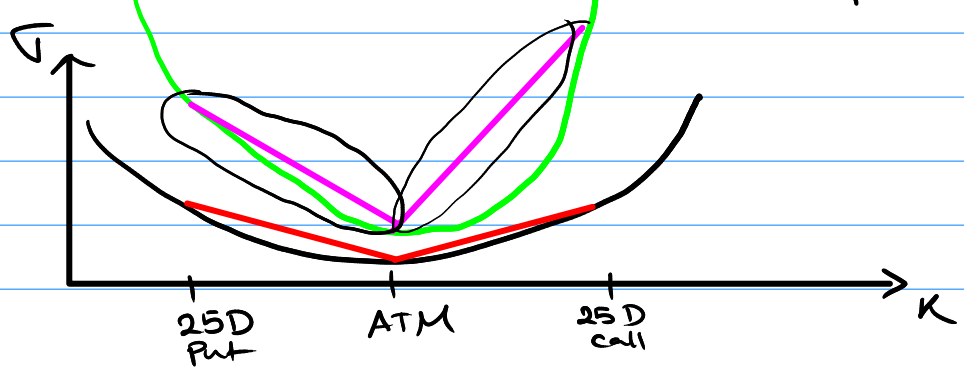
El strike K_2 tal que $\Delta C(K_2) = 0.75$

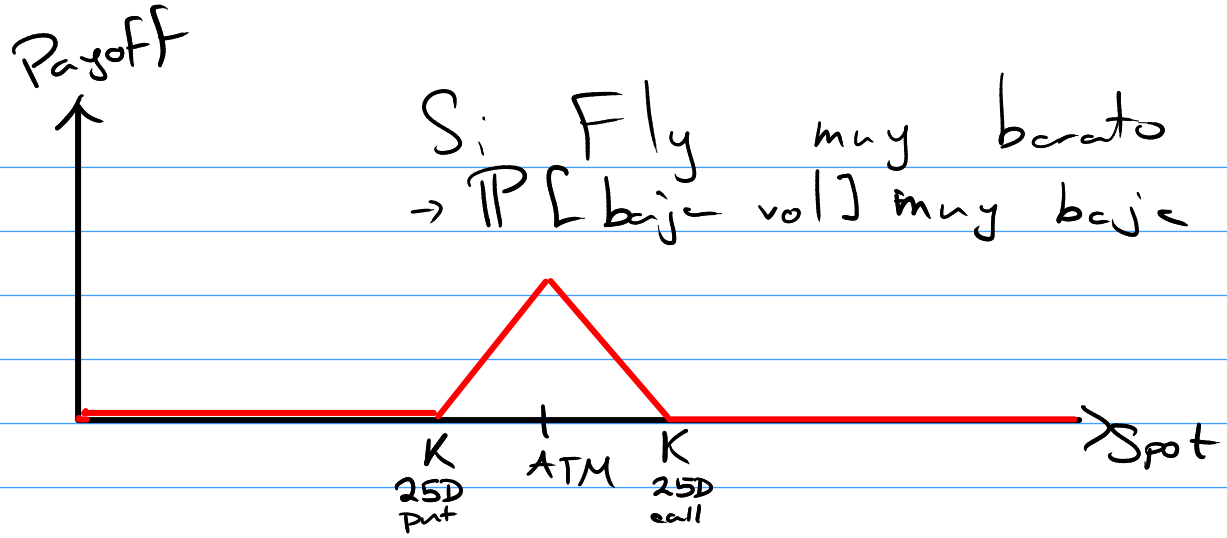
↳ RR mul grande

USD MXN → USD se aprecie en el futuro vs el MXN } Estrategia Direccional

→ Butterfly: $(\sigma_{25D\ call} - \sigma_{ATM}) - (\sigma_{ATM} - \sigma_{25D\ put})$

Fly: $\sigma_{25D\ call} - 2\sigma_{ATM} + \sigma_{25D\ put}$

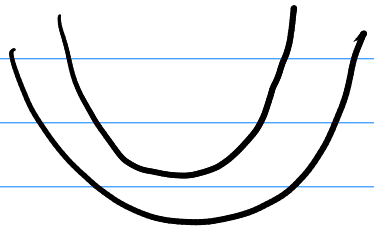




Si Fly muy barato
$\rightarrow \mathbb{P}[\text{baja vol}]$ muy baja

Si Fly es muy caro:
$\rightarrow \mathbb{P}[\text{baja vol}]$ muy alta

$$\sigma(K) = \alpha K + \beta K^2 + \gamma$$



$\alpha, \beta, K \in \mathbb{R}$

Método de Newton:

$\sigma_0 = \sigma_{init}$ $\quad C$ = market price

ε = tolerancia (10^{-8}) $\quad f(\sigma) \rightarrow$ ecuación de BS

mientras $(C - f(\sigma_n))^2 > \text{Epsilon}$:

$$\sigma_{n+1} = \sigma_n - \frac{f(\sigma_n)}{f'(\sigma_n)}$$

$$f'(\sigma_n) = \frac{\partial f(\sigma_n)}{\partial \sigma}$$

↙ Vega

$$\boxed{\sigma^* = \sigma_{n+1}}$$

Calcular la vol. implícita para la fórmula de Black //

$$C_t = D(t,T)\left[F(t,T)N(d_1) - KN(d_2)\right]$$

$$d_1 = \frac{\ln\left(\frac{F}{K}\right) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}\,; \quad d_2 = d_1 - \sigma\sqrt{T}$$

→ Cálculo del forward.

Beneficios de la fórmula de Black

* Separamos los factores corresp. al mercado de Fwds de los factores del mercado de opciones.
* Fórmula muy versátil para cualquier tipo de suby.

→ En eq: $F(t,T) = S_t e^{(r-r_\delta)(T-t)}$

↳ r_δ → dividend yield

→ En FX = $F(t,T) = S_t e^{(r-r_{fx})(T-t)}$

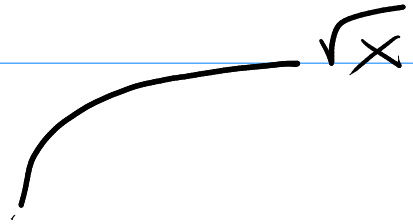
↳ r → tasa dom

r_{fx} → tasa foránea

→ En Comod = $F(t,T) = S_t e^{(r+r_c)(T-t)}$

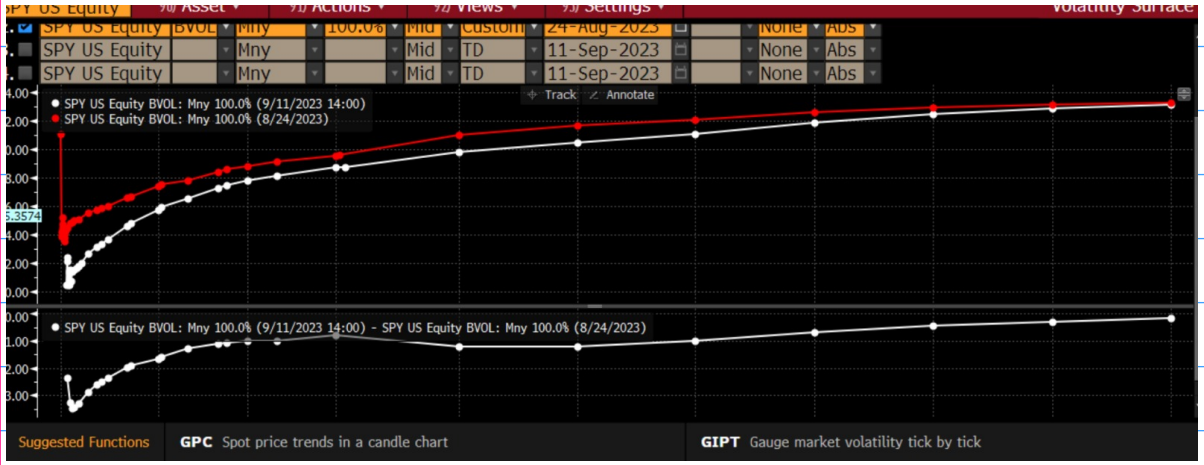
↳ r → tasa desc.

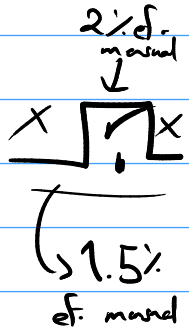
r_c → costos de almacenaje.



→ En tasas $F(t,T,S) = \left[\frac{D(t,T)}{D(t,S)} - 1\right] \cdot \frac{1}{\tau(T,S)}$

Cupón ⇔ Spy ∈ [90%, 110%] respecto al Spot

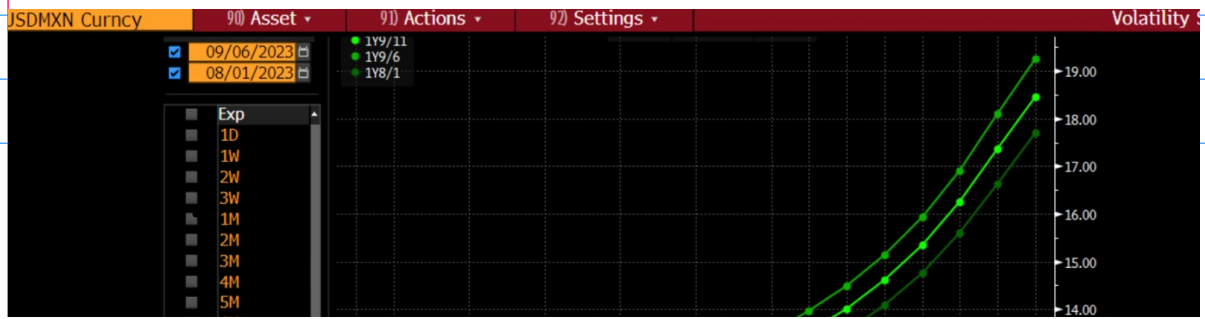


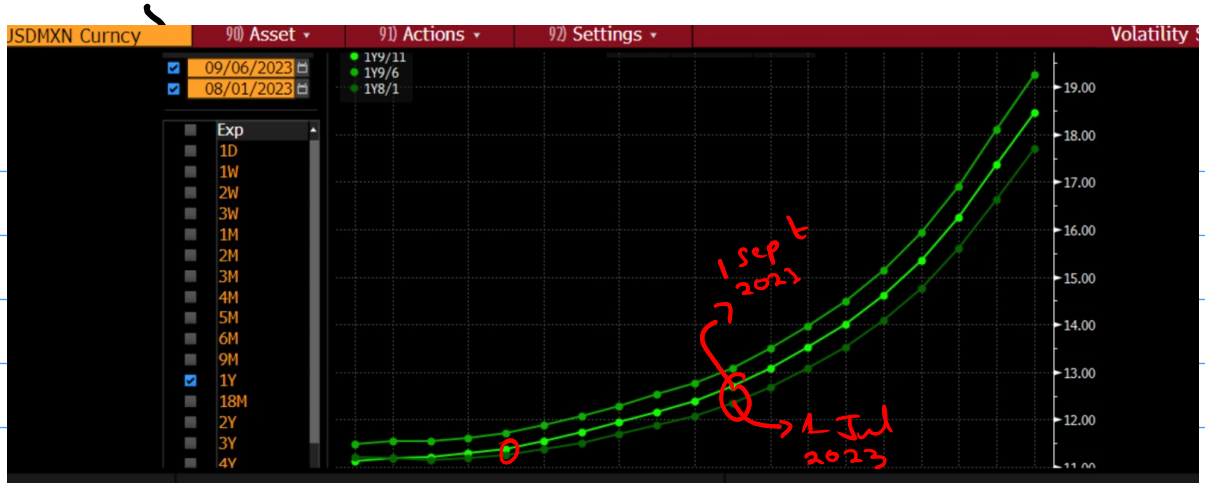




Exp	ATM Bid / Ask	25D RR Bid / Ask	25D BF Bid / Ask	10D RR Bid / Ask	10D BF Bid / Ask
1D	23.105 / 30.210	0.835 / 5.810	-1.455 / 2.100	1.965 / 10.490	-1.660 / 4.025
1W	13.555 / 16.290	2.670 / 4.585	-0.270 / 1.100	5.105 / 8.385	0.450 / 2.640
2W	12.950 / 14.800	3.025 / 4.320	-0.035 / 0.890	5.750 / 7.970	0.880 / 2.360
3W	12.770 / 14.120	3.105 / 4.050	0.110 / 0.785	5.960 / 7.575	1.035 / 2.110
1M	12.850 / 13.800	3.370 / 4.035	0.240 / 0.715	6.600 / 7.740	1.290 / 2.050
2M	12.335 / 13.185	3.460 / 4.055	0.300 / 0.725	6.600 / 7.620	1.395 / 2.075
3M	11.965 / 12.790	3.320 / 3.900	0.320 / 0.730	6.405 / 7.400	1.415 / 2.075
4M	11.945 / 12.740	3.205 / 3.760	0.330 / 0.725	6.160 / 7.115	1.385 / 2.020
5M	11.848 / 12.630	3.154 / 3.697	0.343 / 0.733	6.124 / 7.059	1.413 / 2.036
6M	11.830 / 12.610	3.140 / 3.680	0.340 / 0.730	6.090 / 7.020	1.460 / 2.080
9M	11.950 / 12.795	2.835 / 3.430	0.350 / 0.770	5.640 / 6.655	1.390 / 2.065
1Y	11.965 / 12.745	2.915 / 3.460	0.410 / 0.800	5.650 / 6.585	1.580 / 2.205
18M	12.190 / 13.090	2.790 / 3.420	0.430 / 0.875	5.400 / 6.480	1.540 / 2.260
2Y	12.270 / 13.285	2.795 / 3.500	0.440 / 0.945	5.470 / 6.685	1.620 / 2.430
3Y	12.205 / 13.630	2.620 / 3.615	0.395 / 1.110	5.190 / 6.900	1.635 / 2.775

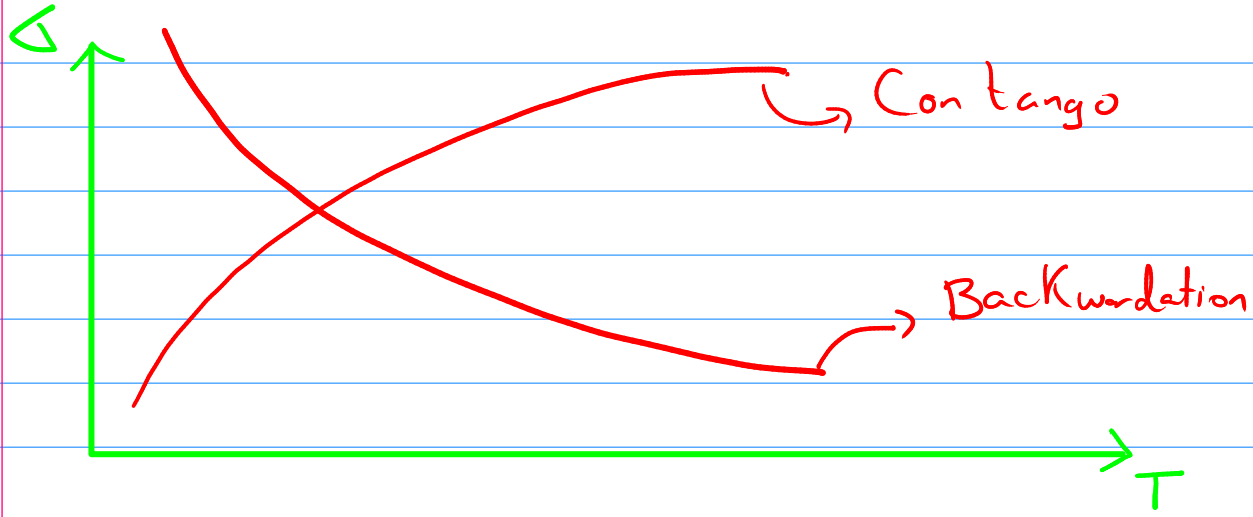





USDMXN Curncy
90) Asset
91) Actions
92) Settings
Volatility
09/06/2023
08/01/2023
1Y9/11
1Y9/6
1Y8/1
Exp
1D
1W
2W
3W
1M
2M
3M
4M
5M
6M
9M
1Y
18M
2Y
3Y
4Y
19.00
18.00
17.00
16.00
15.00
14.00
13.00
12.00
1 Sept 2023
1 Jul 2023
25D Put
ATM
25D Call


97) Settings
98) Output
200) Show in Launchpad
Local Swaps
UDI Swaps
SOFR vs TIIE
BBVA
LOCAL SWAPS
60) MSG Contributor
Zoom
BBVA -> Emerging Markets -> Latam Local Markets -> Mexico -> Swaps -> Local Swaps
BBVA Pays
BBVA Receives
Chg on day
Pay Size (MM)
Receive Size (MM)
1) MXN SWAP MTH (v 28D) 28D 11.4555 11.5055 0.004 1.00M 1.00M
2) MXN SWAP MTH (vs 28D) 3M 11.5200 11.5400 0.010 1.00M 1.00M
3) MXN SWAP MTH (vs 28D) 6M 11.4800 11.5000 0.005 1.00M 1.00M
4) MXN SWAP MTH (vs 28D) 9M 11.4300 11.4500 0.020 1.00M 1.00M
5) MXN SWAP MTH (v 28D) 13M 11.2450 11.2850 0.035 1.00M 1.00M
6) MXN SWAP MTH (v 28D) 26M 10.4000 10.4400 0.100 1.00M 1.00M
7) MXN SWAP MTH (v 28D) 39M 9.8200 9.8800 0.115 1.00M 1.00M
8) MXN SWAP MTH (v 28D) 52M 9.4600 9.5200 0.125 1.00M 1.00M
9) MXN SWAP MTH (v 28D) 65M 9.2800 9.3400 0.135 1.00M 1.00M
10) MXN SWAP MTH (v 28D) 91M 9.2150 9.2750 0.175 1.00M 1.00M
11) MXN SWAP MTH(v 28D) 130M 9.1900 9.2500 0.195 1.00M 1.00M
12) MXN SWAP MTH(v 28D) 195M 9.2700 9.3300 0.195 1.00M 1.00M
13) MXN SWAP MTH(v 28D) 260M 9.2650 9.3250 0.195 1.00M 1.00M
14) MXN SWAP MTH(v 28D) 390M 9.2600 9.3200 0.195 1.00M 1.00M
Cotizacion TIIE vs Fij
13x 1


MPSW1A 11.2650 As of 11 Sep Source BMSW
MPSW1A BMSW Curn
94) Suggested Charts
96) Actions
97) Edit
Line Chart
09/11/2022 - 09/12/2023
BMSW
Last Px
Local CCY
Mov Avgs
Key Events
1D 3D 1M 6M YTD 1Y 5Y Max Daily
Table
+ Related Data
Add Data
Edit Chart
Track
Annotate
News
Zoom
Last Price 11.265
High on 03/08/23 11.91
Average 11.0273
Low on 09/12/22 9.90
11.75%
11.70
SBV
Pagando 11.70
Swap cancelable
11%
25bps → 50 bps
11%
Sep Oct Nov Dec Jan Feb Mar Apr May Jun Jul Aug Sep
2022
2023
Posibilidad de subir


Contango
Backwardation
T

Modelo SABR

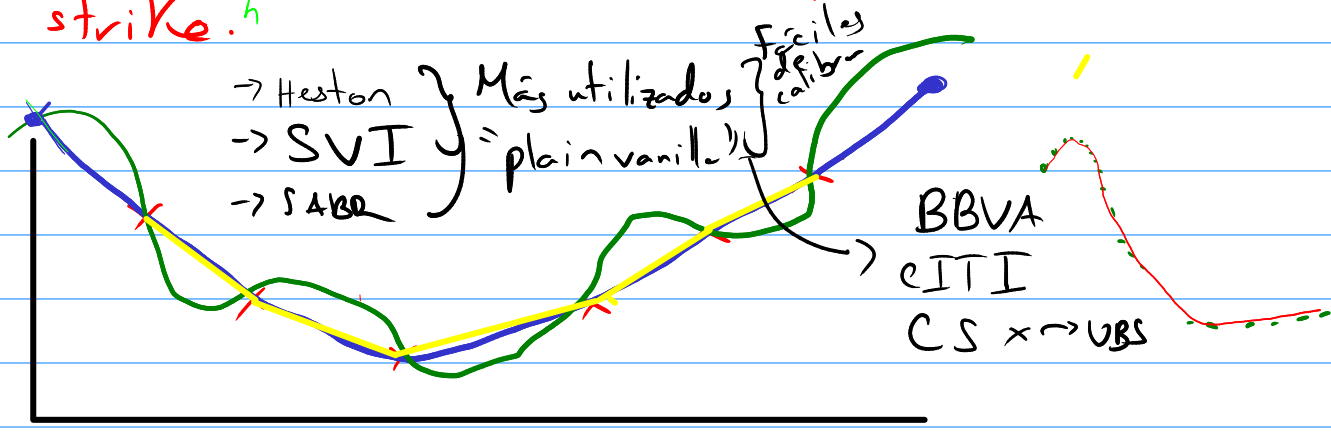
→ "Stochastic alpha, beta, rho, nu"

$$dF(t,T) = \hat{\alpha}_t F(t,T)^{\beta} dW_t \rightsquigarrow F(0,T)$$

$$d\hat{\alpha}_t = \nu \hat{\alpha}_t dZ_t \rightsquigarrow \alpha_0$$

$$dW_t dZ_t = \rho dt$$

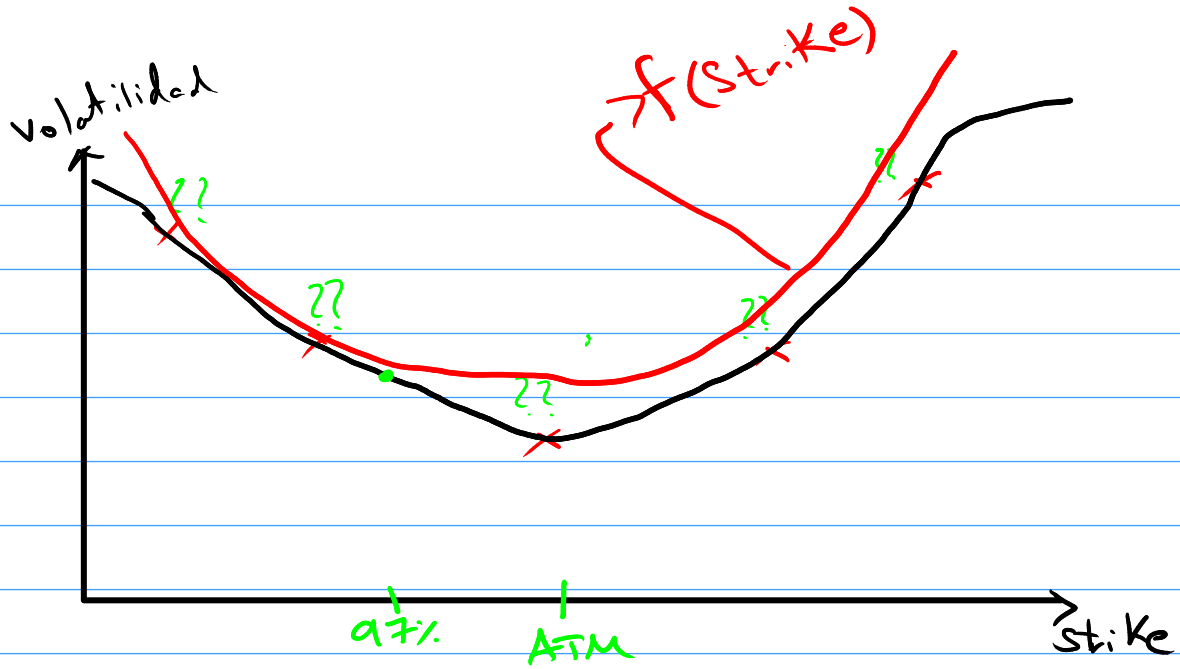
→ El objetivo de la volatilidad implícita de SABR es generalizar la superficie de volatilidad de mercado para cualquier strike.



Modelos "exóticos"

→ Rough Bergoni
→ Fractional Rough Bergoni
⋮

} Difíciles de Calibrar / JP Morgan, GS, M.S.



$$f(K) = \alpha + \beta K + \gamma K^2$$

$Strikes_mkt = \{K_1, K_2, K_3, \ldots, K_n\}$
$vols - mkt = \{\sigma_1, \sigma_2, \sigma_3, \ldots, \sigma_n\}$
} Volatilidades de mercado.

La forma de calibrar la superficie de volatilidad, es encontrando los parámetros (α, β, γ) de mi modelo teórico $(f(K) = \alpha + \beta K + \gamma K^2)$ tal que

$$\rightarrow \min_{(\alpha,\beta,\gamma)} \sum_{i=1}^{n} (f(K_i) - \sigma_i)^2 \text{ es mínima}$$
} Método de mínimos cuadrados.

$$\alpha_1 + \alpha_2 K + \alpha_3 K^2 + \alpha_4 K^3 + \alpha_5 K^4$$

$$\min_{(\alpha,\beta,\gamma)} \sum_{i=1}^{n} |f(K_i) - \sigma_i|$$

$\alpha + \gamma K^2$

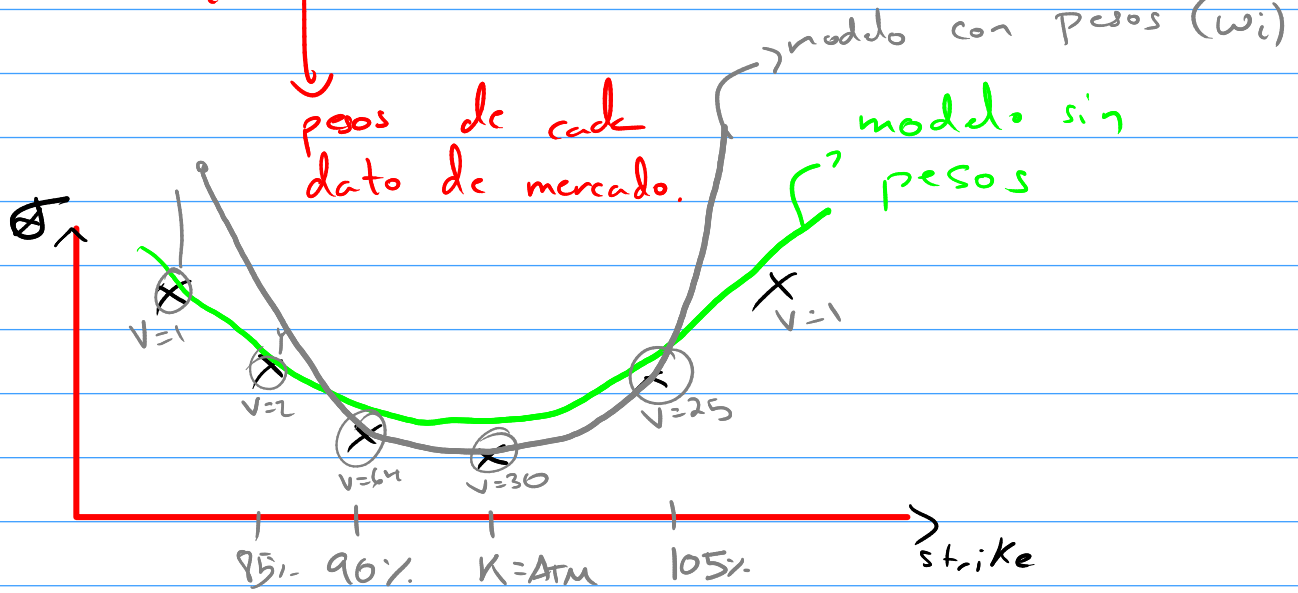
$$\min_{\alpha,\beta,\gamma} \sum_{i=1}^{n} (f(K_i) - \sigma_i)^2 + \lambda(\alpha + \cancel{\beta} + \gamma)^2$$

(λ: parámetro de penalización)

Ayuda a "regularizar" la calibración de la sup. de vol penalizando los parámetros del modelo

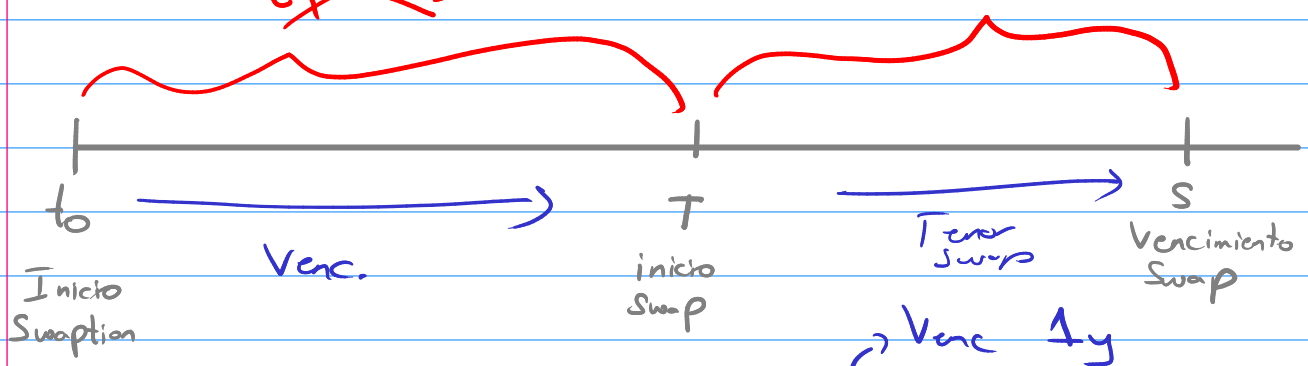
$$\min_{(\alpha,\beta,\delta)} \sum_{i=1}^{n} w_i \left(f(K_i) - \sigma_i\right)^2$$

w_i → pesos de cada dato de mercado.



$\tau_1 = 2y$ ~~$\tau_1 = 1y$~~

$\tau_2 = 3y$ ~~$\tau_2 = 1y$~~



Tenor | Strike

Tenor	99	100	101	102
1y	σ_{11}	σ_{12}	σ_{13}	σ_{14}	σ_{15}
2y	σ_{21}	σ_{22}			σ_{25}
3y	σ_{31}	σ_{32}			σ_{35}

Venc 1y

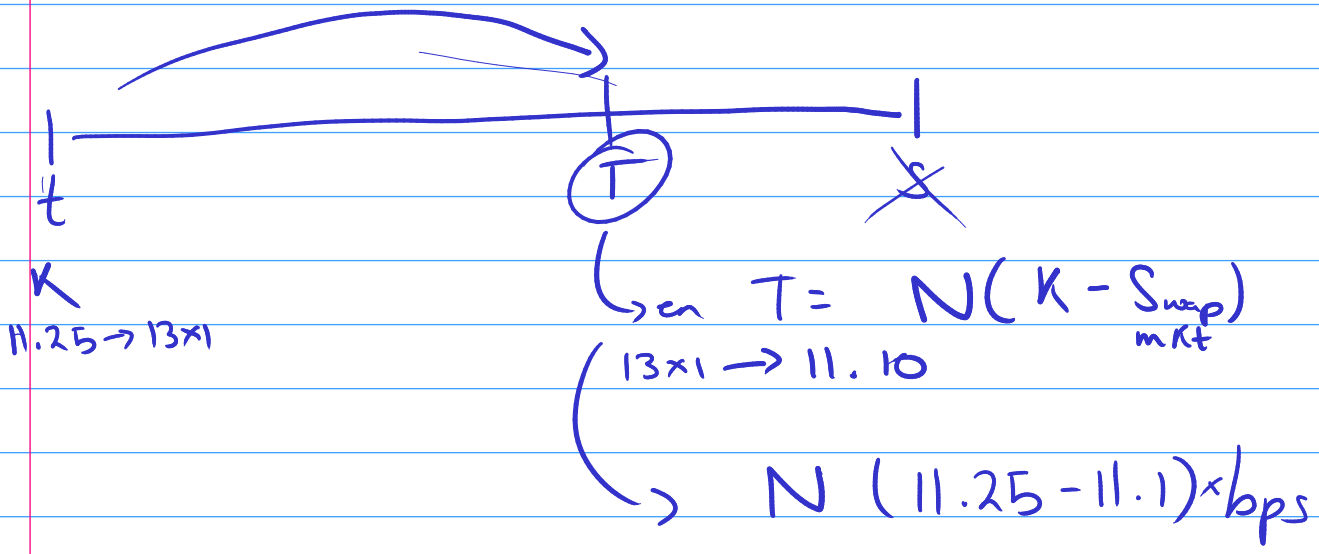
Tenor	K_1	K_2	K_3
1y	σ_{11}	σ_{12}	σ_{13}
2y	σ_{21}	σ_{22}	σ_{23}
3y	σ_{31}	σ_{32}	σ_{33}

Venc. 2y

Tenor	K_1	K_2	K_3
1y	σ_{11}	σ_{12}	σ_{13}
2y	σ_{21}	σ_{22}	σ_{23}
3y	σ_{31}	σ_{32}	σ_{33}

Cubo de volatilidad

Swaption Cash



$$S = \sqrt{\sum_{i=1}^{n} \left(f(K_i, \alpha, \beta, \rho, \upsilon) - \sigma_i \right)^2}$$

f = Sabr

minimizar S

